

%nprocshared=64
%mem=320GB
%chk=PbRhaC10C10T2.chk
opt=loose freq apfd/gen scrf=(solvent=water) pseudo=read scf=xqc

PbRhaC10C10T2

0 1

Pb	-1.58676260	-3.27556662	-0.78546161
O	1.28589297	-0.85887854	-0.33029643
O	-0.95960351	-1.16686784	-1.64516727
O	1.18026695	-3.23001726	-1.81949857
C	4.70958900	-2.87299638	-0.70827814
O	4.73002431	-1.47119174	-1.02499296
C	3.63456620	-0.79567287	-0.43620410
C	2.35475464	-1.13414252	-1.20665056
C	2.44565876	-2.60473069	-1.68003458
C	3.27248697	-3.41712583	-0.69924447
O	-1.16895440	-2.46038258	-3.42857309
C	-0.93948268	-1.35183783	-2.91647082
C	-0.57093174	-0.19348278	-3.81908272
C	-0.71219886	1.21335940	-3.25424769
O	0.26588505	1.45556873	-2.20967778
C	-2.05652733	1.52224563	-2.62479443
C	-3.23994593	1.40531041	-3.57656482
C	-4.55508819	1.61767211	-2.83714501
C	-5.77762986	1.69491079	-3.73839167
C	-7.06759888	1.87366744	-2.94851183
C	1.52189576	1.71165228	-2.57717777
O	1.91930414	1.67066635	-3.72245480
C	2.37839549	2.07301690	-1.39439060
C	3.77994894	1.42844774	-1.41454063
O	3.91772988	0.54696873	-0.29837470
C	5.44892965	-3.16270961	0.58375845
O	3.32317199	-4.78230246	-1.06093492
C	4.87864329	2.46555183	-1.27815349
C	6.27991217	1.87842520	-1.32054524
C	7.36181469	2.94190039	-1.19580195
C	8.77149430	2.36716951	-1.18930366
C	9.85558844	3.43314201	-1.10922132
O	0.16404965	-2.80507353	0.99337991
O	-2.68587339	-1.76236258	0.66197209
O	-0.86844294	-5.09414472	1.68717366
C	-2.23834940	-2.81281656	4.27951705
O	-1.07035082	-1.98356220	4.30823620
C	-0.57743436	-1.67799293	3.01644190
C	0.07153846	-2.92111151	2.40386819
C	-0.75937593	-4.17807730	2.76485464
C	-2.18981650	-3.81781091	3.12683044
O	-3.85776090	-2.18449854	-1.15801974

C	-3.72708890	-1.60732644	-0.05211567
C	-4.81089643	-0.69615239	0.47748257
C	-4.31959073	0.72785240	0.71639701
O	-3.43941832	0.72435975	1.86792698
C	-5.46103307	1.69814815	0.95525203
C	-4.99882762	3.14184742	1.08563414
C	-6.16044057	4.11473227	1.23074324
C	-5.71251862	5.56524941	1.34621016
C	-6.87188751	6.54285011	1.48266466
C	-2.21214197	1.22670852	1.70305891
O	-1.81403433	1.74261609	0.68428493
C	-1.35802980	1.09531690	2.93255581
C	0.05738535	0.59096858	2.57125448
O	0.36808459	-0.67363730	3.16601539
C	-3.50672408	-1.98006418	4.23652139
O	-2.92253792	-4.95914682	3.53688797
C	1.13925859	1.55862536	3.00673438
C	2.54638407	1.07050042	2.69520834
C	3.60209739	2.15767394	2.84274651
C	5.01958179	1.62945044	2.66611259
C	6.08349079	2.71701931	2.70804211
H	0.12129114	0.47213218	1.48492521
H	-3.72932070	1.06864946	-0.13390288
H	3.92750637	0.87425897	-2.34770399
H	-0.49817678	1.91701147	-4.06427892
H	5.21761444	-3.35532307	-1.54804317
H	3.50090324	-1.14634977	0.59557021
H	2.27296807	-0.50671564	-2.09831980
H	2.96308811	-2.61689195	-2.64466917
H	2.83122519	-3.32606401	0.30385600
H	0.46517408	-0.35454279	-4.13466541
H	-1.16743050	-0.28144665	-4.72938112
H	-2.19298498	0.86096999	-1.76833300
H	-2.01418504	2.53955234	-2.22299209
H	-3.14327299	2.13818135	-4.38767847
H	-3.25937958	0.41898431	-4.05463565
H	-4.69291784	0.79575283	-2.12512784
H	-4.49232379	2.53479731	-2.23680905
H	-5.66133967	2.52763417	-4.44434252
H	-5.84384180	0.78488255	-4.34917042
H	-7.19716456	1.04452386	-2.28470941
H	-7.01607239	2.77971100	-2.38165197
H	2.46802512	3.16454448	-1.41567179
H	1.86694785	1.80858003	-0.46957796
H	5.42407219	-4.23543943	0.79149940
H	4.98779892	-2.64462981	1.42915380
H	6.49132702	-2.84672939	0.51069001
H	2.41433483	-5.04831398	-1.24983749
H	4.75269834	3.19775188	-2.08432603
H	4.72513231	3.00762811	-0.33622242

H	6.38576194	1.14304248	-0.51673865
H	6.41527044	1.32499294	-2.25911080
H	7.26517426	3.66132246	-2.01985695
H	7.20044585	3.51699164	-0.27506678
H	8.87908048	1.67698764	-0.34254868
H	8.92284813	1.76038803	-2.09204312
H	9.77735232	4.08556326	-1.95368898
H	9.73389386	3.99869547	-0.20908817
H	-2.19300644	-3.38833266	5.20881574
H	-1.39408536	-1.35906763	2.36060420
H	1.08122804	-3.02259324	2.81699643
H	-0.29751803	-4.66767912	3.62967383
H	-2.66560130	-3.37545995	2.24366935
H	-5.63285484	-0.67233179	-0.23886480
H	-5.19165094	-1.11203446	1.41576087
H	-6.14983297	1.61156143	0.10769106
H	-6.02040626	1.38977954	1.84690071
H	-4.32840395	3.24085104	1.94741426
H	-4.40417235	3.41133135	0.20401836
H	-6.83188050	4.00859437	0.36828766
H	-6.75618836	3.84443332	2.11253546
H	-5.04229263	5.67164654	2.20953668
H	-5.11373433	5.83124979	0.46493729
H	-7.50543059	6.46692192	0.62373621
H	-7.43416424	6.30875383	2.36240543
H	-1.28818343	2.09706157	3.36803375
H	-1.83033328	0.45770556	3.67960628
H	-4.38085682	-2.63640243	4.23438311
H	-3.53568096	-1.36933355	3.33144836
H	-3.56858036	-1.32392227	5.10740736
H	-2.85785834	-5.60616940	2.82492578
H	0.95005168	2.50522230	2.48557305
H	1.03009399	1.76560756	4.07911173
H	2.79260515	0.22660188	3.34836275
H	2.57370676	0.68686199	1.67215003
H	3.41230696	2.94639884	2.10184202
H	3.50809421	2.63887541	3.82610400
H	5.22846691	0.88655134	3.44796853
H	5.08222311	1.09813708	1.71083449
H	5.90449776	3.42152621	1.92284688
H	6.04425061	3.21812441	3.65263402
H	0.72461625	-2.04077151	0.71176461
H	0.00291878	-5.44588354	1.47611076
H	0.87786853	-3.15391411	-2.73223760
H	0.45685466	-0.78993711	-0.86005646
C	-8.25954882	1.94372712	-3.92110990
H	-8.12998271	2.77287065	-4.58491232
H	-8.31107553	1.03768359	-4.48796976
C	-9.56381660	2.12446525	-3.12247401
H	-9.50858756	3.02580154	-2.54850306

H	-10.39100376	2.18237490	-3.79871942
H	-9.69807011	1.29079701	-2.46530336
C	-6.32338147	7.97779442	1.59071994
H	-5.72842299	8.06651523	2.47562278
H	-5.72162980	8.19513785	0.73307389
C	-7.49724350	8.97223443	1.65954574
H	-7.11622365	9.96836054	1.74589780
H	-8.08582001	8.89139238	0.76963436
H	-8.10529193	8.74789218	2.51092559
C	7.47407889	2.08227464	2.52105560
H	7.65307150	1.37776763	3.30625080
H	7.51331929	1.58116959	1.57646366
C	8.55057397	3.18271019	2.56348122
H	8.50631451	3.68883664	3.50516967
H	9.51735456	2.74081352	2.44115934
H	8.37600690	3.88306842	1.77359220
C	11.23990260	2.75840239	-1.10833457
H	11.31813830	2.10598104	-0.26386695
H	11.36159742	2.19284897	-2.00846771
C	12.33646566	3.83663608	-1.02733110
H	13.29799702	3.36734328	-1.01629821
H	12.26471564	4.48286026	-1.87712297
H	12.20858172	4.40886675	-0.13229046

C H O 0
6-311+g(2d,p)

Pb	0	
S	4 1.00	
	591.61124370	0.22126521076D-03
	46.757232559	0.56961959130D-02
	20.746462696	-0.21374063831
	14.610796419	0.40502620616
S	2 1.00	
	20.222637612	-0.83541883299D-01
	6.4767324865	0.97910892388
S	1 1.00	
	1.6600600927	1.0000000
S	1 1.00	
	0.80431655001	1.0000000
S	1 1.00	
	0.22627039020	1.0000000
S	1 1.00	
	0.84014530665D-01	1.0000000
P	3 1.00	
	15.189102118	0.61952303583
	14.693144415	-0.72498497086
	6.8705890048	0.37680007984
P	3 1.00	
	2.2021426123	0.40196284806

		1.2209125119	0.46058131862
		0.63367559815	0.19367655397
P	1	1.00	
		0.28202837058	1.0000000
P	1	1.00	
		0.11333375666	1.0000000
P	1	1.00	
		0.43948707430D-01	1.0000000
D	6	1.00	
		61.315369628	0.33870800787D-03
		12.372195840	0.13788683942D-01
		6.9254944983	-0.75979608103D-01
		2.3319539939	0.28113784298
		1.2108730003	0.44474512269
		0.60090478506	0.35326874351
D	1	1.00	
		0.28135869813	1.0000000
D	1	1.00	
		0.11500000000	1.0000000
F	1	1.00	
		1.0000000	1.0000000
F	1	1.00	
		0.2896200	1.0000000

PB	0		
PB-ECP	3	60	
f potential			
	2		
2		3.88751200	12.20989200
2		3.81196300	16.19029100
s-f potential			
	4		
2		12.29630300	281.28549900
2		8.63263400	62.52021700
2		3.88751200	-12.20989200
2		3.81196300	-16.19029100
p-f potential			
	6		
2		10.24179000	72.27689700
2		8.92417600	144.59108300
2		6.58134200	4.75869300
2		6.25540300	9.94062100
2		3.88751200	-12.20989200
2		3.81196300	-16.19029100
d-f potential			
	6		
2		7.75433600	35.84850700
2		7.72028100	53.72434200
2		4.97026400	10.11525600

2	4.56378900	14.83373100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100